# RAKESH CHOWDARY POLINA

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## PROFILE

Software Engineer with 4+ years of experience in building scalable web applications, RESTful APIs, and database-driven solutions. Proficient in Java, Python, PHP (CodeIgniter), SQL, and AWS, with expertise in microservices, facial recognition, and CI/CD automation. Successfully optimized payroll systems, implemented machine learning pipelines, and enhanced application security, reducing downtime and improving performance. Adept at Agile methodologies, API documentation (Swagger), and DevOps.

## EDUCATION

**Master of Science in Computer Science** **May 2024**
Oklahoma State University, Stillwater, OK **CGPA: 3.909/4**

## SKILLS

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| **Languages:** | PHP, Java, JavaScript, Python, SQL, C, C++, HTML, CSS |
| **Frameworks:** | React JS, CodeIgniter, Spring boot, Node JS, Hibernate, Spring MVC |
| **Databases:** | MySQL, MongoDB, NoSQL, PostgreSQL, Cassandra |
| **Web Technologies:** | RESTful API, Spring JDBC, JPA, OAuth 2, jQuery, JSP, JMS, RabbitMQ |
| **Web Services:** | REST, SOAP |
| **Tools/Platforms:** | Docker, Git, JIRA, Jenkins, Postman, Ansible, Kubernetes, Apache Kafka, Terraform, Swagger |
| **AWS Cloud Services:** | EC2, S3, EKS, RDS, ELB, Cloud Watch, CloudFront |
| **Build Tools:** | Maven, Gradle |

## WORK EXPERIENCE

***Research Data Analyst, Oklahoma State University*** **May 2023 – May 2024**

- Engineered and implemented a robust facial recognition pipeline, performing both quantitative and qualitative analysis on image data with Python and InsightFace, to achieve a precision rate of 93.6%
- Implemented DBSCAN clustering and rigorously verified data accuracy across 182,000+ images, enhancing archive reliability and insight precision.
- Designed an intuitive UI for the labeling tool using Flask and React.js, streamlining user engagement and annotations.
- Designed and maintained database structures using SQL, enhancing user interaction management and feedback data integrity, ensuring operational excellence.

***Software Engineer, Ericsson, India*** **July 2019 – July 2022**

- Transitioned legacy monolithic HR and payroll web applications to modular PHP-based architectures using CodeIgniter framework, improving scalability by decomposing monolithic workflows into reusable modules. Integrated asynchronous task queues (RabbitMQ) for payroll batch processing, reducing peak-time server load by 35%.
- Developed RESTful APIs with CodeIgniter for benefits enrollment and employee data management systems, adhering to OpenAPI specifications. Implemented role-based access control (RBAC) and input validation to address security vulnerabilities, reducing downtime by 25% and mitigating SQL injection risks.
- Designed and optimized relational databases (MySQL) for HR workflows, including payroll calculation and employee leave management. Wrote complex SQL queries with JOINs and stored procedures, improving report generation speed by 40% for large datasets.
- Collaborated with HR stakeholders to analyze requirements for a global employee self-service portal. Prototyped UI components using CodeIgniter's MVC structure and integrated with SAP ERP via SOAP APIs, streamlining leave request approvals by 30%.
- Automated CI/CD pipelines using Jenkins and Git for PHP deployments, reducing manual errors by 50%. Containerized payroll modules with Docker for consistent testing environments and deployed to AWS EC2 instances, cutting deployment time by 40%.
- Implemented unit and integration testing with PHPUnit and Postman, achieving 85% test coverage for critical payroll APIs. Conducted security audits using OWASP guidelines, resolving 15+ vulnerabilities related to session management and data encryption.
- Migrated legacy employee databases to a normalized structure, reducing redundancy by 60%. Utilized CodeIgniter's Query Builder for efficient data operations and implemented database indexing, accelerating search queries by 25%.
- Collaborated with cross-functional teams using Agile/Scrum, leading bi-weekly sprint planning for HR application upgrades. Documented APIs with Swagger and trained support teams on issue tracking (Jira), reducing onboarding time by 30%.
- Optimized AWS infrastructure for a benefits enrollment system using RDS, S3, and Elastic Load Balancer. Reduced server costs by 20% through auto-scaling groups and reserved instance planning.

## TECHNICAL PROJECTS

***Sleep data analysis*** **January 2024**

- Applied LSTM network and Granger causality techniques to investigate sleep data on a dataset comprising over 5,000 individuals' sleep patterns, unveiling insights into temporal dynamics and causal connections within the dataset.